

Course syllabus for

Academic year 2024/2025 >

TIF285 - Learning from data

Inläring från data

Course syllabus adopted 2024-01-31 by Head of Programme (or corresponding)

 Owner: [MPPHS](#)
7,5 Credits
Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: Second-cycle

Main field of study: Engineering Physics

Department: 16 - PHYSICS

Course round 1

Teaching language: English

Application code: 85114

Open for exchange students: Yes

Block schedule: [C+](#)

Maximum participants: 60 (at least 10% of the places are reserved for exchange students)

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course [!]	No Sp	Examination dates [!]
0119 Project 7,5 c Grading: TH	7,5 c						

In programs

[MPPHS PHYSICS, MSC PROGR, Year 1 \(compulsory\)](#)
Examiner:
[Christian Forssén](#)
 [Go to course homepage](#)

Eligibility

General entry requirements for Master's level (second cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

English 6 (or by other approved means with the equivalent proficiency level)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Course specific prerequisites

- Solid background in undergraduate mathematics (multivariable analysis, linear algebra, mathematical statistics)
- Programming skills (the Python programming language will be used throughout the course)
- General physics knowledge (to understand the context of the scientific examples that will be used)

Aim

The course aims to give a deeper theoretical understanding, and in practice experience, of workflows and methods that are essential for performing scientific modeling, statistical inference, and machine learning. Much emphasis is put on probabilistic approaches within science and engineering, such as the ability to quantify the strength of inductive inference from prior knowledge and experimental data to scientific hypotheses and models.

The course is project-based, and the students will be exposed to fundamental research problems and development tasks, with the aim to reproduce state-of-the-art scientific results. The students will use the Python programming language, with relevant open-source libraries, and will learn to develop and structure both workflows and computer codes for scientific modeling and data analysis projects.

Learning outcomes (after completion of the course the student should be able to)

- plan and perform scientific data analysis with methods from Bayesian statistics.
- simulate multivariate probability distributions with MCMC methods.
- quantify and critically assess uncertainties of model parameters via statistical inference.
- understand and numerically implement several probabilistic algorithms used in data analysis and machine learning.
- address open questions in scientific data analysis and perform numerical studies using Python as a programming language.
- write well-structured technical reports where results and conclusions from a scientific data analysis are communicated in a clear way.
- maintain a scientific and ethical conduct in the process of modeling, analyzing data and writing computer programs.

Content

The course has two central parts:

1. Bayesian inference and workflows in scientific modeling
2. (Probabilistic) machine learning methods

The following subtopics will be covered

- Statistical models
- Bayesian statistics and methodology
- MCMC simulation
- Non-linear models and optimization
- Gaussian processes
- (Bayesian) neural networks

Organisation

- Lectures
- Supervised group work on numerical projects
- Analytical and numerical homework exercises
- Computational projects with written reports

Literature

Course compendium.

Additional reading:

Phil Gregory, Bayesian Logical Data Analysis for the Physical Sciences, Cambridge University Press, 2010

D.S. Sivia, Data analysis---a Bayesian tutorial, Oxford Science Publications, 2006

David J.C. MacKay, Information Theory, Inference, and Learning Algorithms, Cambridge University Press, 4th printing, 2005

Trevor Hastie, Robert Tibshirani, Jerome H. Friedman, The Elements of Statistical Learning, Springer, 2nd edition, 2009

Andrew Gelman et al, Bayesian Data Analysis, CRC Press, 3rd edition, 2014

Aurelien Geron, Hands-On Machine Learning with Scikit-Learn and TensorFlow, O'Reilly, 1st edition, 2017

Examination including compulsory elements

The final grade is based on the graded performance on homework assignments (exercises) and computer projects.

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers on educational support due to disability.